YOUR LAPTOP — everything below runs locally & privately

OUTSIDE (leaves the laptop)

app.py — "Hexa Query" UI
dark mode · chat history · Stop · charts

run_pipeline.py — CLI
scheduled / one-shot (n8n)

TWO MODES

MODE 1 · Full Analysis
pipeline.py — autonomous
DQ -> failures -> fixes -> verify -> reports

MODE 2 · Chat
chat_agent.py — intent router
database · web · conversation

Groq cloud LLM
llama-3.3-70b
writes SQL only
never sees a result row

schema + question only

SHARED ENGINE (hexa_analyst package)

TIER 1 · CURATED — trusted anchor numbers

dq_checks.py
YOUR DQ SQL

failure_checks.py
YOUR failure SQL

groundedness.py
grade fix vs sources

report.py
heatmap HTML

TIER 2 · GENERATIVE — leads to verify

sql_agent.py
LangGraph + self-repair

semantic.py
22 metric definitions

egress.py
the only way out

charts.py
auto visual + drill-down

db.py · guard.py (SELECT-only, v1.13.5) · scrub.py (Presidio) · providers.py (cloud->local failover) · config.py

Public search
via SearXNG proxy
generic keywords only

keywords only

LOCAL SERVICES (on your machine)

Ollama
qwen2.5 LLM + nomic embed
ALWAYS explains the results

SearXNG
in Docker · :8080
self-hosted search

SSH tunnel
localhost -> read-only DB
SELECT-only DB user

read-only

Hexa Postgres
remote VPS (host redacted)
READ-ONLY · never written

OUTPUTS

Two .html reports
Comparison + Action Plan · fixes flagged ✓ / ⚠

Chat answer + SQL + chart
number first · then why · hover & drill down